



Attempt to Measure the Unmeasurable

ucal — the Universe Calendar, in two pages

The irritation

Recombination occurred about 380,000 years after the Big Bang. A good sentence, and its units are a problem. A year is the time Earth takes to circle the Sun — definitionally, not approximately. So an event 13.8 billion years old is described in units defined by a planet that would not exist for another nine billion. The number is correct; it carries a passenger.

Not a complaint about cosmology. The complaint is that **provenance leaks into arithmetic**: compute in Earth years and Earth's orbital period sits inside every intermediate value.

Three things are declared. Everything else in the system is **computed** from them — the notation, the ladder, the domain, the calendars, the cosmology. They are worth the space.

1 — The tick

The Planck time, $t_P = \sqrt{\hbar G/c^5} \approx 5.391247 \times 10^{-44}$ s. It is the one combination of the gravitational constant, the reduced Planck constant and the speed of light with the dimension of time — so it is built from the bounds on gravitation, quantum action and causality, and from nothing else. It does not know about Earth, or planets, or matter being organised at all.

Every quantity in the system is an unsigned integer count of ticks. Nothing else is primitive — not the second, not the beat, not the day.

It is not a quantum of time. It is the resolution floor of an instrument, not a structure found in the world. The system cannot represent a shorter duration; that is a fact about the system. Asserting otherwise would take a side in a twenty-four-century argument as a side effect of choosing an integer type.

The one concession. The tick's length in **seconds** is fixed by convention against SI, because stating it at all needs a unit to state it in. That concedes metrology and nothing else: ticks are counted, never converted.

2 — The beat

5^{60} ticks — 867 361 737 988 403 547 205 962 240 695 953 369 140 625 of them, about 46.762 ms. The specification calls it the **universe second**.

Two choices, each with one reason. **Base five**, because $5^5 = 3125$ is a number of exactly five base-5 digits, so a tier is a clean five-digit group. **Exponent 60**, because it puts the reference rung at the scale a human can notice a duration — the one concession to the reader in the whole design, and it changes no arithmetic.

Everything above and below is 5^{5k} : deep, drift, span, sweep, arc, beat, flicker. A uniform ladder, no ragged fields at either end.

The beat is not a second in disguise. One second is 21.385061835 beats, and no rescaling would make it whole: **SECOND** carries thirty factors of five and **BEAT** carries sixty, so the two share a common measure only **at the tick**.

That is the whole reason the tick is primitive rather than either of them.

3 — The datum

Tick 0. **Stipulated** — declared, not measured, and not an observed event.

It cannot be measured, and the reason is arithmetic before it is philosophy. The published age is 13.787 ± 0.020 Gyr; that uncertainty is a fifty-eight-digit number of ticks, 0.145% of the whole span. Define the datum **as** the measured age and every timestamp inherits the wobble, making the exact arithmetic theatre.

So it is declared: **ORIGIN_OFFSET** = 9 304 311 741 502 590 385 beats, a whole number of them. The published age rounded to a whole beat — and the system prints what the rounding discarded, -0.017190364 s, rather than absorbing it.

Two things follow. The domain is **unsigned**: it begins at the datum, and an earlier instant is not representable — **UCAL-E0020**, an error rather than a negative number. A limit on what can be **dated**, not a claim about what exists; the specification keeps the origin claim in a **signed** window precisely because the limit may lie earlier. And the physical claim about where the origin falls is kept separately — cited, with its exact magnitude, in a type with **no arithmetic operations at all**. Three tests exist whose job is to **fail to build** if anyone computes with it.

What is computed from them

Nothing below is a further decision. Each follows from the three above.

The notation. A timestamp is the tick count written in base 5 and grouped in fives. So truncation **is** rounding, prefix comparison **is** chronological comparison, and writing every digit pinpoints one tick — no separate rounding step, no scaling, no drift between the coarse view and the fine one.

The domain. 512 bits, reaching 2.29×10^{103} years — past proton decay and black-hole evaporation. The present epoch is 6×10^{-94} of it. The width is fixed so that it never has to change: the canonical binary form is 64 bytes because the domain is 512 bits.

The bridge. Earth enters through one exact constant, **SECOND** = 18 548 584 399 861 $\times 10^{30}$ ticks. In is multiplication and never rounds; out is division and is the only place a rounding mode is chosen. The dependency is not eliminated — that is not available to anyone — it is **localised** to somewhere you can point at.

Calendars. A body's periods enter as exact rationals of ticks; intercalation is **derived** by continued fraction, never declared.

Cosmology. No floating point anywhere, so ages are certified interval quadrature over exact rationals: two numbers and a proof the answer lies between them, rather than one number and a guess.

The philosophical background

The line between what a measurement **establishes** and what it merely **points at** is old, and it does not stay where you put it. Kant was bluntest: the illusion that erodes it is natural, unavoidable, and **persists after diagnosis** — the astronomer knows the moon is no larger at the horizon and goes on seeing it so.

The usual remedy is vigilance, which lasts as long as attention does. In 1922 Florensky — who had the distinction available in his own tradition, fifteen centuries deep — computed a radius from the Lorentz factor outside its domain of validity and identified the resulting surface with the Empyrean. One move, and nothing in his method forbade it.

Kant's First Antinomy supplies the reason the origin cannot be measured at all: both **the world has a beginning and it does not** are demonstrable, and both fail, because both assume a completed totality about which the question has an answer waiting. Stipulating is not a retreat from a better method. There is no better method.

Cusanus, 1440: no proportion holds between finite and infinite, so precision does not constitute approach.

The theological background

Basil of Caesarea, fourth century: the beginning of a road is not the road. A beginning is not a member of the series it begins — which is the datum's content, with no arithmetic.

Gregory of Nyssa makes **διώστημα**, interval, the mark of createness: everything created has it, and what lies beyond has none. Every quantity in this system is an interval, so the title stops being a paradox and becomes a statement about categories.

The Hebrew calendar's epoch is **molad tohu**, "the new moon of chaos" — computational, placed before the event it anchors, and named for its own emptiness. This project spent four chapters explaining that its epoch is stipulated; that one puts it in the name. Its **helek** was chosen for divisibility exactly as **SECOND** was.

And a validity window — assert a parameter where it was measured, warn beyond — is al-Ghazālī's **āda** with the theology removed.

None of this argues that any tradition is correct. The decisions have older, better-named precedents reached for other purposes; reporting that is an argument that the decisions are not eccentric, and nothing more.

What came out of it

Give the mechanism nothing but Earth's rotation and orbit and its first answer is **1/4** — the Julian calendar, with no knowledge of Rome. **97/400** never appears: the Gregorian rule is not a convergent at any depth, and **31/128** is **124× more accurate** with a denominator three times smaller.

The specification claimed twice, in print, that the mechanism reproduces both. It reproduces one. That correction is published, and a build-time assertion now stops it being reversed.

Two more fell out. The Metonic cycle — 235 months in 19 years, known to Babylon — appears from two numbers unaided. And Mars has **no month**: neither moon is one, and the mechanism returns nothing rather than inventing one, because **month-like** is an Earth predicate.

What is claimed, and what is not

A measuring instrument may legitimately point at what it cannot describe, provided it declares that it is only pointing — and that declaration can be enforced mechanically rather than left to the author's discipline.

The first clause is Cusanus'. The second is the contribution: a comment addresses a reader who is paying attention, a runtime check addresses a program that reaches a line, and a **type** addresses everyone who ever writes against the library — including someone who disagrees. They are not persuaded. They are refused, in under a second.

Not claimed: that time began; that the Big Bang happened at tick 0; that the tick is time's smallest unit; that base 5 means anything; that any tradition is right; or that the system is **useful** — no task you have needs a Planck-tick count, and that is the second of three facts rather than an apology.

Where next. **cargo install ucal**, or **cargo add ucal-core**. The normative specification is **spec/UCAL-1.1.md**; the twenty-four rules and what enforces each are in **spec/RULES.md**. A 29-page account is **UCAL_INTRO**. The full argument — nine traditions, six samples, and a chapter on what they failed to establish — is **Life, the Universe, and God**, 251 pages.